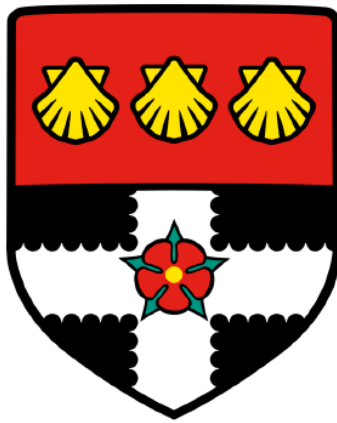


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DEPARTMENT OF METEOROLOGY



Improving Surface Gravity Wave Forecasts Using Data Assimilation

Owen Aljabar

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*A thesis submitted in partial fulfillment of the requirements for the degree of
Master of Science in Atmosphere, Ocean and Climate*

August 2024

Declaration of Authorship

I, Owen Aljabar, declare that this thesis titled, “Improving Surface Gravity Wave Forecasts Using Data Assimilation” and the work presented in it are my own. I confirm that:

- This work was done wholly or mainly while in candidature for a research degree at this University.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.
- Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

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Abstract

Faculty Name

Department of Meteorology

Master of Science

Improving Surface Gravity Wave Forecasts Using Data Assimilation

by Owen Aljabar

The Thesis Abstract is written here (and usually kept to just this page). The page is kept centered vertically so can expand into the blank space above the title too...

Acknowledgements

The acknowledgments and the people to thank go here, don't forget to include your project advisor. . .

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List of Abbreviations

DA	Data Assimilation
WW3	WAVEWATCH III
LETKF	Local Ensemble Transform Kalman Filter

Chapter 1

Introduction

1.1 The need for Reliable Wave Forecasting

Offshore operations form an important of many industries such as energy and research, and contribute to the economies of coastal countries round the world. Ocean waves pose a significant risk to offshore operations as they can lead to the loss of life and damage to equipment. This means that accurate wave predictions are a critical safety issue for offshore operations, the the operations can only be carried out safely when the waves are sufficiently calm. To provide these predictions the MetOcean Forecasting division of Fugro produces wave forecasts for a client undertaking offshore operations in the Gulf of Mexico. To avoid unnecessary downtime, they produce ensemble forecasts and apply corrections using a proprietary method based on the alpha factor method from (cite).

(Description of site specific alpha factor)

In this project, we explore the potential of Data Assimilation (DA) to reduce uncertainty and error in these wave forecasts. We will also examine the ensemble statistics, which are an important component of ensemble-based DA systems.

1.2 Previous work on DA for Wave Forecasts

Houghton et al., 2023 assimilated wave height observations from satellite altimeters and a global network of drifting Sofar spotter buoys. They compared two different DA methods, a deterministic optimal interpolation (OI) method and an ensemble-based local ensemble transform Kalman filter (LETKF). They found the both methods reduced forecast error out to 2.5 days leadtime, and the LETKF gave more physically realistic output states. In this project we will assimilate observations of both wave height and period in the hope that correlations between these variables might give better predictions of both. We will use the ensemble transform Kalman filter (ETKF).

Houghton et al., 2022 used OI to compare the effects of assimilating wave height and period. They found that both variables give very similar improvements in predictions of wave height, while spectral assimilation gives better results for other quantities such as period and direction.

Caires, Kim, and Groeneweg, 2018 investigated the effect on wave forecasts of using an ensemble Kalman filter (EnKF) method to adjust the wind fields used as forcing in a wave model. They found...

1.3 Main Research Questions

1: What are the statistical relations in an ensemble of wave height and period forecasts?

We will examine forecasts from Fugro of significant wave height H_s , along with two measures of the peak period T_{m01} and T_{m02} (defined in equation 3.4). We will look at the statistics most relevant to ensemble DA systems, including:

- Ensemble spread
- Spatial covariances of each variable
- Cross covariances between variables

These covariances are particularly important in DA as they determine how information from the observations is spread around the domain.

2: What kind of reduction in forecast error can be achieved by assimilating H_s , T_{m01} and T_{m02} using the ETKF method?

We will use ETKF to assimilate observations of the three variables and examine the nature of the forecast errors after DA. We will look at...

Chapter 2

Theory

Here some lines need to be added informing the reader about what this chapter covers and what the aims are of this chapter.

2.1 Linear Wave Theory

This section is too short. In my view three topics should be discussed. 1) A short qualitative discussion of the processes that create and destroy waves and influence their propagation backed up by references. 2) The linear wave theory for propagation on which WWIII is based. This is already largely there. 3) The link between linear wave theory and the energy spectra that are modelled of WWIII. Make sure to link those spectra to the quantities we are working with (Hs, Tm01, Tm02).

Ocean waves are a distinctive visual feature of the ocean surface, and very important to offshore operations such as wind farms. They are an instance of surface gravity waves. This section will outline the **linear wave theory**

It should be made clear that this is an approximation. Higher-order wave theories like Stokes and cnoidal waves exist, but are treated here as WWIII is based on linear wave theory.

relevant to this project, as derived in (Vallis, 2017, page 253), ~~page 253~~.

Parameters describing surface gravity waves include the wavenumber k ~~and~~, direction θ , and the frequency ω . The first two describe the wave vector **k**

You better create a separate command for vectors and matrices. This ensures that you don't have to plough through the whole text if you want to change the format for vectors

, which has magnitude **k** and direction θ

Add $\mathbf{k} = k \cos \theta \hat{\mathbf{x}} + k \sin \theta \hat{\mathbf{y}}$ in which you clearly define what are directions of $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ as well as the definition of θ

. ω here denotes the absolute frequency measured in a fixed frame of reference. If the medium is subject to currents, this will differ from the relative frequency σ measured

in a frame of reference moving with the mean current, due to the d Doppler effect. This difference is quantified by the equation

$$\omega = \sigma + \mathbf{k} \cdot \mathbf{U} \quad (2.1)$$

and σ and k are related by the dispersion relation

$$\sigma^2 = gk \tanh kD \quad (2.2)$$

The dispersion relation has two limiting forms:

- Deep water waves, where $kD \gg 1$, so $\tanh kD \approx 1$ and $\sigma^2 \approx gk$.
- Shallow water waves, where $kD \ll 1$, so $\tanh kD \approx kD$ and $\sigma^2 \approx gk^2D$

The phase speed is then given by

$$c = \frac{\sigma}{k} \approx \begin{cases} \sqrt{\frac{g}{k}} & \text{in deep water} \\ \sqrt{gD} & \text{in shallow water} \end{cases} \quad (2.3)$$

and the group speed

Explain the physical relevance of the group speed vs the phase speed

may be expressed as

$$c_g = \frac{\partial \sigma}{\partial k} = \frac{1}{2}c \left(1 + \frac{2kD}{\sinh 2kD} \right) \approx \begin{cases} \frac{1}{2}c & \text{in deep water} \\ c & \text{in shallow water} \end{cases} \quad (2.4)$$

We can now use scale analysis to see that ω and σ generally have similar values. The ratio between $\mathbf{k} \cdot \mathbf{U}$ and σ in equation 2.1 is

$$\frac{\mathbf{k} \cdot \mathbf{U}}{\sigma} \sim \frac{kU}{\sigma} = \frac{U}{c}$$

Where $U = |\mathbf{U}|$ is the mean current speed. Since $c \gg U$ in general, this ratio is small and so $\omega \approx \sigma$.

Note that equations 2.1 - 2.4 are valid for linear waves of a single frequency and direction. Real ocean waves are more complicated than this, containing multiple frequencies propagating in multiple directions. The WAVEWATCH III model used in this project models ocean waves as a superposition of these simple linear waves.

Surface waves with multiple frequencies and directions are represented using a wave spectrum $F = F(\sigma, \theta)$, which is also a function of position and time. F represents the wave energy associated with a particular frequency σ and direction θ , so the total wave energy at any point in space and time is

$$E = \iint F d\sigma d\theta \quad (2.5)$$

Here, we also define the wave spectral moments which are derived from the unidirectional spectrum $S(\sigma) = \int F(\sigma, \theta) d\theta$:

$$m_k = \int S(\sigma) \sigma^k d\sigma \quad (2.6)$$

These moments can be used to derive other important wave variables, such as significant wave height and dominant wave period, from the spectrum (expand). Finally we define the wave action

This better moves to the paragraphs on linear wave theory.

$$N(\sigma, \theta) = \frac{F(\sigma, \theta)}{\sigma}$$

N is conserved in the absence of sources and sinks, making it the spectrum of choice for wave models.

2.2 Data Assimilation

Strictly speaking we are using the update step of the ETKF which is somewhat unusually. This section can be used to show that this approach is in absence of localization and with a linear operator is equivalent to the weak 4DEnVar.

As well as the model, we also have some direct observations of the waves which we would like to make use of to improve our predictions. We use techniques of data assimilation to achieve this.

Add general explanation what DA is doing. I.e. finding estimates that lie closer to truth using the information in the observations.

We take a vector $\mathbf{x} \in \mathbb{R}^n$ to represent the state of the system

Extend the description of how $\mathbf{x}$ is related to the model

, assuming it has a normal background distribution $\mathbf{x} \sim N(\mathbf{x}_b, \mathbf{B})$

. We take a vector $\mathbf{y} \in \mathbb{R}^p$ to represent a set of observations, and assume $\mathbf{y}$ is related to $\mathbf{x}$ via an observation operator $\mathcal{H} : \mathbb{R}^n \rightarrow \mathbb{R}^p$:

It might be better to introduce $\mathbf{x}_{true}$ here to make a distinction between a specific value of the state and reference to the state in general.

$$\mathbf{y} = \mathcal{H}(\mathbf{x}) + \varepsilon$$

where the error term ε is assumed to be normally distributed with mean zero and covariance $\mathbf{R}$: $\varepsilon \sim N(0, \mathbf{R})$. We call the vector spaces containing $\mathbf{x}$ and $\mathbf{y}$ the state

space and observation space respectively. In operational contexts, the dimensions of these spaces are very large: typically $n \sim 10^8$ and $p \sim 10^6$. This means that considerable computational resources are required in most cases and computational efficiency is a key consideration in any operational DA implementation.

In our case $\mathcal{H}$ is linear i.e. there is a matrix $\mathbf{H} \in \mathbb{R}^{n \times p}$ such that $\mathcal{H}(\mathbf{x}) = \mathbf{H}\mathbf{x}$ and so $\mathbb{P}(\mathbf{y}|\mathbf{x}) = N(\mathbf{H}\mathbf{x}, \mathbf{R})$. We can then express the distribution of $\mathbf{x}$ given some set of observations $\mathbf{y}$ using Bayes' theorem:

$$\mathbb{P}(\mathbf{x}|\mathbf{y}) = \mathbb{P}(\mathbf{y}|\mathbf{x}) \frac{\mathbb{P}(\mathbf{x})}{\mathbb{P}(\mathbf{y})} \quad (2.7)$$

This allows $\mathbb{P}(\mathbf{x}|\mathbf{y})$ to be determined since $\mathbb{P}(\mathbf{y}|\mathbf{x})$ and $\mathbb{P}(\mathbf{x})$ are known, and $\mathbb{P}(\mathbf{y})$ is independent of $\mathbf{x}$ so it acts as a normalisation constant. Since all the distributions involved are normal, $\mathbb{P}(\mathbf{x}|\mathbf{y})$ will be normally distributed also:

You need define $\sim$ here.

$$\mathbb{P}(\mathbf{x}|\mathbf{y}) \sim e^{-\frac{1}{2}(\|\mathbf{x} - \mathbf{x}_b\|_{\mathbf{B}^{-1}}^2 + \|\mathbf{y} - \mathbf{H}\mathbf{x}\|_{\mathbf{R}^{-1}}^2)} \quad (2.8)$$

adopting the notation $\|\mathbf{a}\|_{\mathbf{M}}^2 = \mathbf{a}^\top \mathbf{M} \mathbf{a}$. Finding the most likely value of $\mathbf{x}$ given this distribution is equivalent to minimising a cost function

$$J(\mathbf{x}) = \frac{1}{2} \|\mathbf{x} - \mathbf{x}_b\|_{\mathbf{B}^{-1}}^2 + \frac{1}{2} \|\mathbf{y} - \mathbf{H}\mathbf{x}\|_{\mathbf{R}^{-1}}^2 \quad (2.9)$$

The state that minimises this cost function is called the analysis $\mathbf{x}_a$ and may be expressed as

reference needed and d_b^o should be a bold vector

$$\mathbf{x}_a = \mathbf{x}_b + \mathbf{K}d_b^o \quad (2.10)$$

where $d_b^o = \mathbf{y} - \mathbf{H}\mathbf{x}$ is the innovation and $\mathbf{K} = \mathbf{B}\mathbf{H}^\top (\mathbf{H}\mathbf{B}\mathbf{H}^\top + \mathbf{R})^{-1}$ is the Kalman gain. Note that the increment $d_b^a = \mathbf{x}_a - \mathbf{x}_b$ is given by $\mathbf{B} \left(\mathbf{H}^\top (\mathbf{H}\mathbf{B}\mathbf{H}^\top + \mathbf{R})^{-1} d_b^o \right)$, so it always lies in the column space of $\mathbf{B}$. This means that $\mathbf{B}$ is crucial to determining how information from the observations is spread around the domain by the DA system. In particular, if $\mathbf{B}$ is rank deficient, then all possible increments lie within a strict subspace of the state space. (so?)

Limits the degrees of freedom of the DA corrections.

I don't think the remaining part of this section is relevant as there is no prediction step in our setup. Introduction of the ensemble estimate of $\mathbf{B}$, the analysis mean and analysis covariance would be beneficial at this point.

In most contexts, observations are not all simultaneous **assimilated** but instead come from a sequence of times $t = 0, 1, \dots$. One approach to assimilating non-simultaneous observations is to use a filter method such as the Ensemble Kalman Filter (Anderson, 2001; Evensen, 1994; Houtekamer and Mitchell, 1998). A filter method consists of successive analysis steps, in which the observations from time t are assimilated according to equation 2.10, alternating with update steps, in which the forecast model is used to evolve the state forward to time $t + 1$.

In the ensemble Kalman filter (EnKF), a forecast ensemble is used and each ensemble member is evolved separately by the filter. The background error covariance matrix $\mathbf{B}$ is approximated at each analysis step by the forecast error covariance matrix

$$\mathbf{P}_t^f = \frac{1}{N-1} \sum_{i=1}^N \left(\mathbf{x}_t^{(i)} - \bar{\mathbf{x}}_t \right) \left(\mathbf{x}_t^{(i)} - \bar{\mathbf{x}}_t \right)^\top$$

Where N is the ensemble size, $\mathbf{x}_t^{(i)}$ is the state of ensemble member i before the analysis step at time t and $\bar{\mathbf{x}}_t = \frac{1}{N} \sum_{i=1}^N \mathbf{x}_t^{(i)}$ i.e. the ensemble mean.

In general, the observations from each time t may be available at different locations, or observations of different variables may be available, so the observation operator and observation error covariance matrix are functions of time. Denoting these at time t by $\mathbf{H}_t, \mathbf{R}_t$, the Kalman gain matrix takes the form

$$\mathbf{K}_t = \mathbf{P}_t^f \mathbf{H}_t^\top \left(\mathbf{H}_t \mathbf{P}_t^f \mathbf{H}_t^\top + \mathbf{R}_t \right)^{-1}$$

The EnKF method is prone to a problem called filter divergence - sampling error in the calculation of $\mathbf{P}_t^f$ means the magnitude of the forecast error covariance tends to erroneously shrink over time (Jazwinski, 1970). This means that after some time, the ensemble distribution becomes very tight and further observations do not affect the state of the ensemble, resulting in forecast analysis states which diverge from the true state. (schematic) To counteract this, we will employ the method of inflation described in Anderson and Anderson, 1999, where the forecast error covariance matrix is multiplied by a factor α before the analysis step is performed.

2.2.1 (Description of 4DEnVar - may not be needed)

In our case, we have observations from multiple times $t = 0, 1, \dots, T$ spread out over some time window, and we expect the state $\mathbf{x}$ to evolve over this time window. This evolution may be predicted using the model as follows: let $\mathbf{x}_0$ represent the state of the system at $t = 0$, then we define the model operator $\mathcal{M}_t$ for each t such that $\mathbf{x}_{t+1} = \mathcal{M}_t(\mathbf{x}_t)$ as predicted by the model. The cost function then becomes

$$J(\mathbf{x}_0) = \frac{1}{2} \|\mathbf{x}_0 - \mathbf{x}_b\|_{\mathbf{B}_0^{-1}}^2 + \frac{1}{2} \sum_{t=0}^T \|\mathbf{y}_t - \mathbf{H}_t \mathbf{x}_t\|_{\mathbf{R}_t^{-1}}^2 \quad (2.11)$$

$$= \frac{1}{2} \|\mathbf{x}_0 - \mathbf{x}_b\|_{\mathbf{B}_0^{-1}}^2 + \frac{1}{2} \sum_{t=0}^T \|\mathbf{y}_t - \hat{\mathbf{h}}_t \mathbf{x}_0\|_{\mathbf{R}_t^{-1}}^2 \quad (2.12)$$

where the observations $\mathbf{y}_t$ and matrices $\mathbf{H}_t, \mathbf{R}_t$ are valid at time t , and

$\hat{\mathbf{h}}_t$ is an operator and should therefore be capitalized

$\hat{\mathbf{h}}_t = \mathbf{H}_t \mathcal{M}_{0 \rightarrow t}$ using the abbreviation $\mathcal{M}_{0 \rightarrow t} = \mathcal{M}_{t-1} \mathcal{M}_{t-2} \dots \mathcal{M}_0$.

We will use the 4DEnVar method, where a control variable transform is used to write $\mathbf{x}_0$ as a linear combination of the ensemble perturbations:

$$\begin{aligned}\bar{\mathbf{x}}_b &= \frac{1}{N} \sum_{n=1}^N \mathbf{x}_b^{(n)} \\ \mathbf{X}'_b &= \frac{1}{\sqrt{N-1}} \left(\mathbf{x}_b^{(1)} - \bar{\mathbf{x}}_b, \mathbf{x}_b^{(2)} - \bar{\mathbf{x}}_b, \dots, \mathbf{x}_b^{(N)} - \bar{\mathbf{x}}_b \right) \\ \mathbf{x}_0 &= \bar{\mathbf{x}}_b + \mathbf{X}'_b \chi\end{aligned}$$

Here $\mathbf{x}_b^{(n)}$ represents the background state from ensemble member n , and N is the ensemble size. The cost function can then be rewritten in terms of χ and simplified by linearising $\hat{\mathbf{h}}_t$ around $\bar{\mathbf{x}}_b$:

$$J(\chi) = \frac{1}{2} \|\chi\|^2 + \frac{1}{2} \sum_{t=0}^T \left\| \mathbf{y}_t - \hat{\mathbf{h}}_t \bar{\mathbf{x}}_b - \hat{\mathbf{H}}_t \mathbf{X}'_b \chi \right\|_{\mathbf{R}_t^{-1}}^2 \quad (2.13)$$

where $\hat{\mathbf{H}}_t = \mathbf{H}_t \mathbf{M}_{0 \rightarrow t}$ is the linearisation of $\hat{\mathbf{h}}_t$. This cost function is simplified by taking a background error covariance matrix of the form $\mathbf{B} = \mathbf{X}'_b \mathbf{X}_b'^\top$, which reduces to the identity under the control variable transform. This cost function has gradient

$$\nabla J(\chi) = \chi + \sum_{t=0}^T \left(\hat{\mathbf{H}}_t \mathbf{X}'_b \right)^\top \mathbf{R}_t^{-1} \left(\hat{\mathbf{h}}_t \mathbf{X}'_b \chi + \hat{\mathbf{H}}_t \bar{\mathbf{x}}_b - \mathbf{y}_t \right)$$

A major advantage of this formulation is that we can make the approximation

$$\mathbf{M}_{0 \rightarrow t} \left(\mathbf{x}_b^{(n)} - \bar{\mathbf{x}}_b \right) \approx \mathcal{M}_{0 \rightarrow t} \left(\mathbf{x}_b^{(n)} \right) - \overline{\mathcal{M}_{0 \rightarrow t} \left(\mathbf{x}_b \right)}$$

Since $\mathbf{H}_t \mathbf{M}_{0 \rightarrow t} \left(\mathbf{x}_b^{(n)} - \bar{\mathbf{x}}_b \right)$ is the n^{th} column of $\hat{\mathbf{H}}_t \mathbf{X}'_b$, this approximation allows us to compute the cost function and its gradient without a tangent linear or adjoint model, which is a computationally expensive aspect of most variational data assimilation schemes (Bannister, 2017).

Chapter 3

Experimental Setup

3.1 Model Description and Domain

Waves are modelled by Fugro using WAVEWATCH III (WW3), which uses as its base spectrum a wavenumber-direction spectrum $F(k, \theta)$, rather than the frequency-direction spectrum described above. The spectrum is discretised using 29 sampled wavenumbers k and 24 angles θ . The wavenumber-direction spectrum is related to the frequency-direction spectrum by the Jacobian transform

$$F(\sigma, \theta) = \frac{\partial k}{\partial \sigma} F(k, \theta) = \frac{1}{c_g} F(k, \theta)$$

Wave propagation is then described by the equation

$$\frac{D}{Dt} N(k, \theta) = \frac{S}{\sigma} \quad (3.1)$$

Where $N(k, \theta) = F(k, \theta)/\sigma$ as before and S describes the net effect of sources and sinks for the spectrum $F(k, \theta)$.

In deep water, S consists of three main parts: an atmospheric interaction term S_{in} , a nonlinear wave-wave interaction term S_{nl} and a dissipation term S_{ds} .

S_{in} is dominated by an exponential wind-wave growth term, so S_{in} is equal to some coefficient multiplied by $\sigma N(k, \theta)$. This leads to a feedback which is usually positive but can be negative in the case of swell, and represents the main source of wave energy.

S_{ds} is generally dominated by wave breaking, and it represents a sink of wave energy. Similar to S_{in} , it is equal to a negative coefficient multiplied by $\sigma N(k, \theta)$. The balance between S_{in} and S_{ds} determines the overall growth or decay of wave energy.

S_{nl} represents interactions between waves due to nonlinear processes, which results in energy transfer between waves of different frequencies. (maybe expand - Janssen, 2004)

In shallow water, processes such as wave-bottom interactions, wave breaking and reflection off shorelines become important, so additional source terms are included to describe this.

From the forecasted spectra, other variables such as wave height, period and mean direction are derived. The three variables we will be focusing on in this study are significant wave height H_s , and two measures of the dominant wave period T_{m01} and

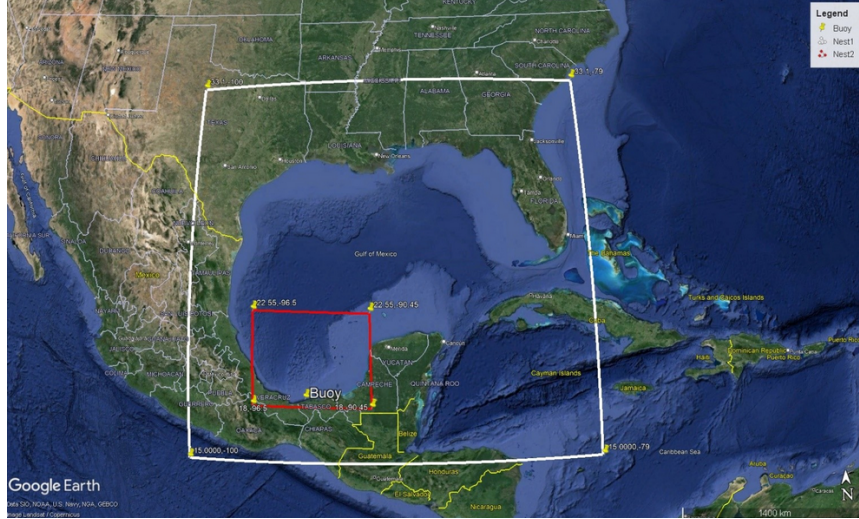


FIGURE 3.1: Model domains shown on map.

Tm02. These are related to the spectrum via the spectral moments defined in 2.6 as follows:

$$H_s = 4\sqrt{m_0} \quad (3.2)$$

$$Tm01 = \frac{m_0}{m_1} \quad (3.3)$$

$$Tm02 = \sqrt{\frac{m_0}{m_2}} \quad (3.4)$$

As forcing, WW3 uses just the 10 metre wind, and this comes from the GEFS ensemble. The GEFS forecast contains one control run plus 20 perturbed ensemble members. Each GEFS run acts as forcing for one member of the wave ensemble. (Do they?)

The domain in which forecasts are produced lies between $15.0 - 33.0^\circ\text{N}$ and $79.1 - 100^\circ\text{W}$ in the gulf of Mexico. Fugro use a nested domain to produce the forecasts, running WW3 on a 0.5° resolution globally rectangular grid, and two higher resolution nested grids covering the region of interest. There is a 0.1° resolution grid spanning the region described above, and a 0.05° grid spanning from $18 - 22.5^\circ\text{N}$ and $90.5 - 96.5^\circ\text{W}$. These domains are shown on a map in figure 3.1.

Forecasts are produced at 12Z each day, and run up to seven days ahead. We have seven of these daily runs available, with start dates spanning from 06/03/2022 to 12/03/2022. (include schematic) As we do not have access to the model code in this project, we will treat each of these runs as one 'timestep' of the DA system. Further description of how this works can be found in section 3.4.

The available model runs give forecast data for a period spanning from 12Z on 06/03/2022 to 19/03/2022. The main feature of this period is a large swell event with north-northwesterly waves, associated with a cold front passing through the domain. This event is illustrated in figure 3.2.

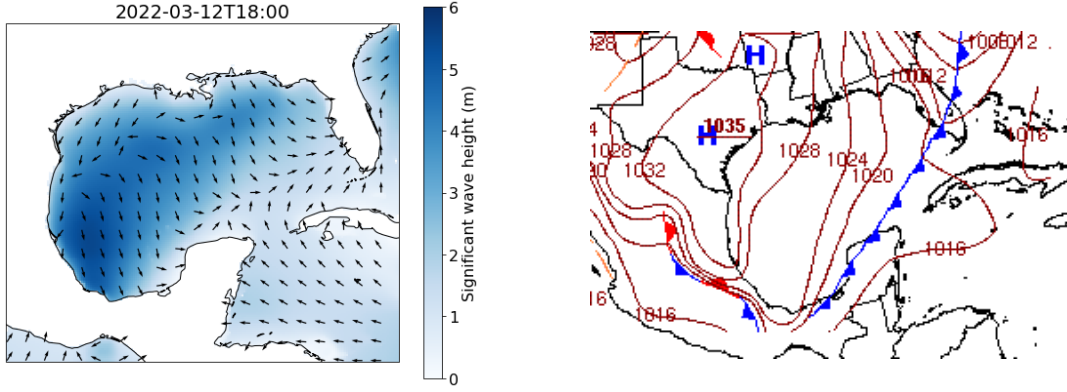


FIGURE 3.2: Left: Significant wave height (colours) and direction (arrows) in the domain at 18Z on 12/03/2022. Right: Surface analysis chart valid for 18Z on 12/03/2022, taken from NOAA WPC surface analysis archive

3.2 ADCP Observations

Our observations come from an acoustic doppler current profiler (ACDP) buoy, which estimates the spectrum by measuring the wave induced velocity and surface height variations. According to Vallis, 2017, the velocity and surface height anomaly induced by a simple wave with wave vector $\mathbf{k} = (k_x, k_y)$, frequency σ and amplitude a is

$$\begin{aligned} u &= a \frac{k_x}{\sigma} g \exp i(\mathbf{k} \cdot \mathbf{x} - \sigma t) \cosh k(z + D) \\ v &= a \frac{k_y}{\sigma} g \exp i(\mathbf{k} \cdot \mathbf{x} - \sigma t) \cosh k(z + D) \\ w &= -ia \frac{k}{\sigma} g \exp i(\mathbf{k} \cdot \mathbf{x} - \sigma t) \sinh k(z + D) \\ \eta &= a \exp i(\mathbf{k} \cdot \mathbf{x} - \sigma t) \end{aligned}$$

where only the real part of each expression should be taken. Note that the horizontal velocity (u, v) is always in the same direction as the wave vector $\mathbf{k}$. From the ADCP measurements, a timeseries of wave induced energy per unit area $E = \int_{-D}^{\eta} \frac{1}{2} \rho_0 \|\mathbf{u}\|^2 dz + \frac{1}{2} \rho_0 \eta^2$ can be calculated. This is equivalent/a good approximation to the wave energy in equation 2.5 (source) so a Fourier transform of E gives the wave spectrum integrated across all directions $E(\sigma) = \int F(\sigma, \theta) d\theta$ (how is directional distribution estimated?)

Other variables are then deduced from this spectrum, including Hs, Tm01 and Tm02 as described above.

We have observations available over a roughly 6 month period spanning from October 2021 to April 2022, which includes the period for which forecasts are available. A timeseries of the observed significant wave height from the forecast period is included in figure 3.3. The peak in wave height late on the 12th of March is clearly visible on the plot.

To determine the magnitude of the observation errors, several methods are available. A simple one is to plot a histogram of observation - ensemble mean, called an $o - b$

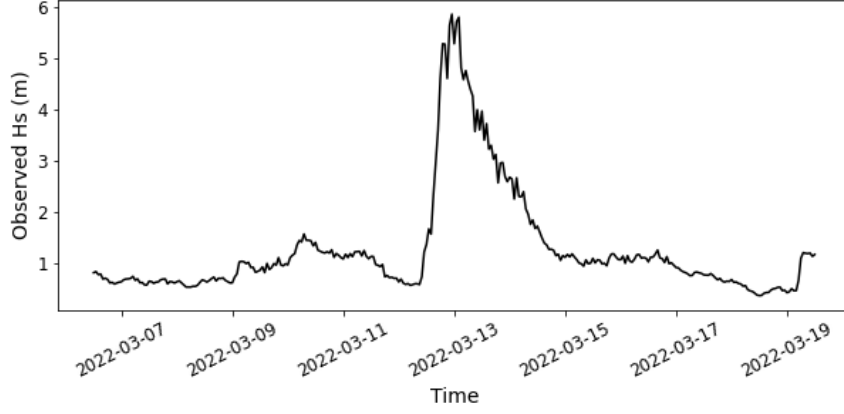


FIGURE 3.3: Observed timeseries of significant wave height from ADCP buoy

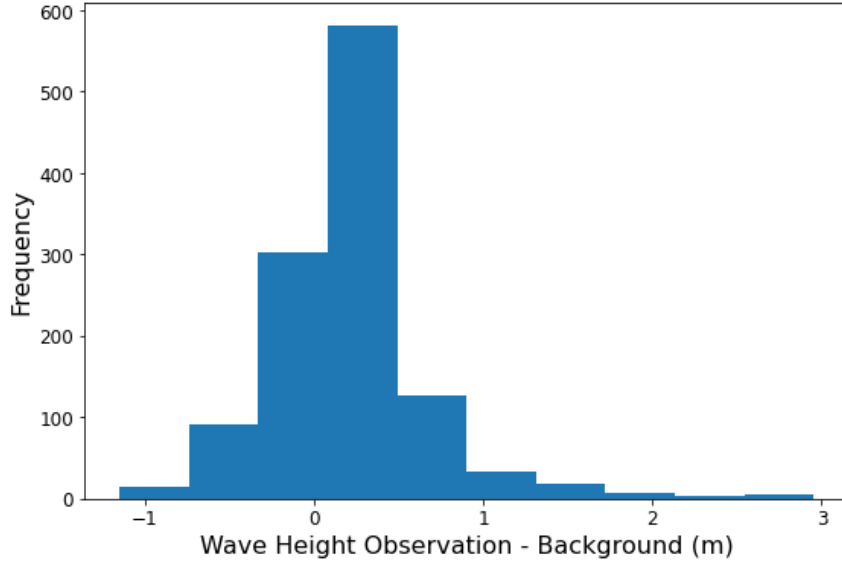


FIGURE 3.4: Observation - background histogram for significant wave height.

plot. The spread in $o - b$ comes from a combination of observationa and background error, so a simple way of estimating the observation error is to assume that observation error accounts for a constant proportion of this error, say 50%. An $o - b$ histogram for significant wave height is shown in figure 3.4. From the plot we see that the spread on the histogram is approximately 1m, so the standard of observational errors in significant wave height may be estimated to be approximately 50cm.

This approach may be quantified more precisely following Desroziers et al., 2006, by defining $d_b^o = \mathbf{y} - \mathbf{H}\mathbf{x}_b$ and using the relation

$$\mathbb{E} \left[d_b^o (d_b^o)^\top \right] = \mathbf{H} \mathbf{P}^f \mathbf{H}^\top + \mathbf{R}$$

Taking traces and dividing by the number of observations, we get a good estimate for the observation error variance which is

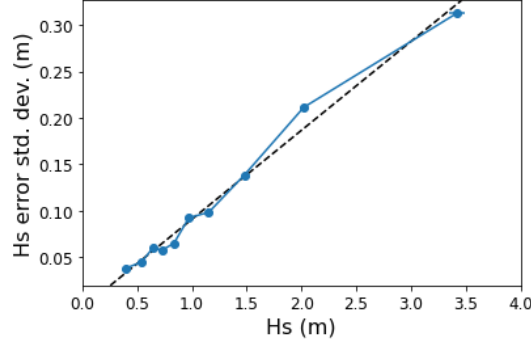


FIGURE 3.5: A plot of Hs error standard deviation against Hs.

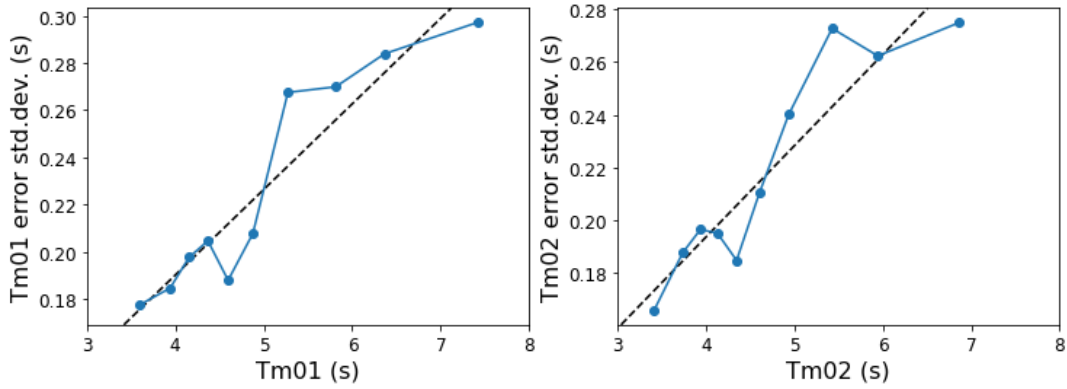


FIGURE 3.6: Plots of Tm01 and Tm02 error standard deviation against Tm01 and Tm02 respectively.

$$\frac{\mathbb{E} \left[\sum_i (d_b^o)_i^2 \right] - \text{Tr} (\mathbf{H} \mathbf{P}^f \mathbf{H}^\top)}{\# \text{ observations}}$$

In our case, the observation operator is very simple and simply selects the forecast values from the appropriate location and times, so the diagonal elements of $\mathbf{H} \mathbf{P}^f \mathbf{H}^\top$ are simply the ensemble forecast variances at the observation location and times.

Another available method is that used in Houghton et al., 2023. They used a collocation study to estimate the error in wave height observations from a distributed network of drifting sofar spotter buoys. Pairs of simultaneous observations from locations within 50km of each other were collected and binned according to the average wave height of each pair. The difference of each pair was taken, and the standard deviation of differences from each bin was computed. They found a strong dependence of error on wave height.

In our case, we only have observations from one buoy and they are available hourly, so we take the pairs to be observations one hour apart. We then calculate the standard deviation using the method described above to obtain the plot in figure 3.5. As in Houghton et al., 2023, we find a strong linear relationship between errors and wave height, with the error standard deviation being approximately 9.6% of the wave height. Repeating this process with the periods Tm01 and Tm02 yields the plots shown in figure 3.6.

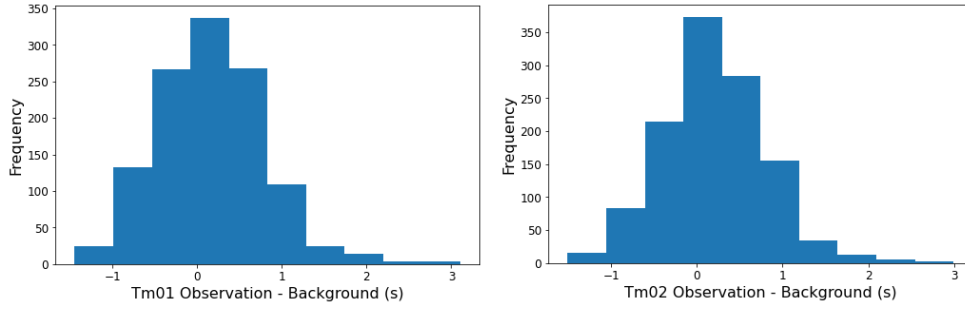


FIGURE 3.7: Observation minus background histograms for Tm01 (left) and Tm02 (right).

The error distributions for Tm01 and Tm02 also fall approximately on straight lines, though the fit is not as good as for Hs. The gradients are also shallower, with the error standard deviations being approximately 3.5% of the observed variable for both. So we will model the error standard deviation in Hs as being equal to 10% of Hs, and the error standard deviation in Tm01 and Tm02 as 3.5% of Tm01 and Tm02 respectively. To avoid 'overfitting' the observations of period (explain and cite), I set a minimum threshold of 0.3s (revise) for the period errors and used the scaling method when it gives a value above this threshold, setting the error equal to the threshold otherwise. This threshold was determined from $o - b$ histograms for Tm01 and Tm02, shown in figure 3.7

3.3 Twin Experiment

In this study, we will not be using real observations in our assimilation but setting aside one ensemble member to represent the 'truth' and generating fictional observations based on this forecast data. This is known as a twin experiment, and has several advantages over assimilating real observations:

- The 'truth' is available away from the location of observations, so forecast error can be calculated over the whole model domain. This allows the effectiveness of DA to improve forecasts away from observation locations to be evaluated

The observations are generated by sampling values from the truth at predefined observation locations and times. Here we sample hourly values from the nearest gridpoint to the buoy to match the real observations that are available. To represent observation error, some random noise is added to the sampled values. This noise is drawn from a normal distribution with standard deviation equal to the error standard deviation derived in section 3.2

3.4 Data Assimilation Setup

To perform the data assimilation we use pyPDAF, the python interface for the fortran-based Parallel Data Assimilation Framework (PDAF). (cite Yumeng?) We will use the ETKF method which comes builtin with PDAF. As we are not able to run the WAVEWATCH III model ourselves, at each analysis step we will correct one 7-day forecast window all at once, and then the forecast step will consist of selecting data from the next 7-day window. In this way the time within the 7-day window is treated like another spatial coordinate by the DA system. A schematic of this process is included in figure 3.8

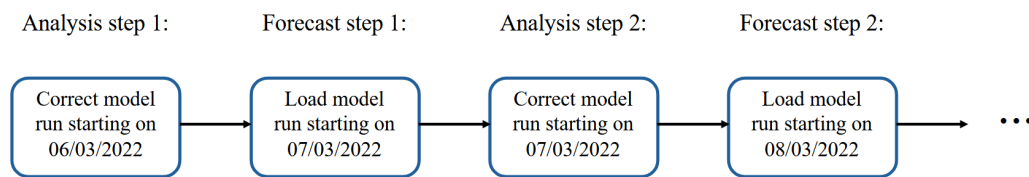


FIGURE 3.8: Schematic of the forecast and analysis steps from the ETKF scheme employed in this project.

At each analysis step, we will assimilate observations from the first three days of the forecast window. This allows us to investigate errors at times after the assimilation window and assess the capability of DA to improve the forecast at future time when observations are unavailable.

Chapter 4

Ensemble Statistics

4.1 Ensemble Spread

The ensemble spread is an important statistic for ensemble based DA systems as it gives us the general scale of the background error covariances. This in comparison with the observation error tells us how much weight will be given to observations at a given location and time. Figure 4.1 shows a timeseries of the ensemble variance in Hs from the run beginning 10th March, alongside the ensemble mean. This is shown both for the buoy location and for a spatial average over the entire domain.

As with the observations, we see that the variance is a strong function of the mean. The variance at the buoy location peaks just before the peak in the mean, and drops back to lower value as the mean reaches its peak. This means that observations from just before the peak will be given higher weighting in the DA than most, as the background error is high compared to the observation error. Conversely, observations taken during and shortly after the peak will be given lower weighting as the observation error at this point is high while the background error is low.

From the plot of spatial mean, we see an overall upward trend in ensemble variance over time, as is to be expected in any forecast. This means that observations closer to the end of the assimilation window will usually be given higher weighting than those taken earlier, since the background error will be greater in comparison to the observation error.

Figure 4.2 Shows the same as figure 4.1 but for Tm01. From the buoy timeseries, we see a sharp peak in variance just before the peak in Tm01, similar to what we saw for Hs. We also see that the swell event is preceded by a dip in Tm01 ensemble mean, which also corresponds with a peak in variance just before. (from this we may hypothesise that variance is highest when the variable in question is changing quickly - test) Since Tm01 observation errors also scale with Tm01, this means that observations from from shortly before the event will be given higher weighting and observations taken during and shortly after will be given lower weighting. The spatial mean plot shows a very strong upward trend in variance with leadtime, so as with Hs, Tm01 observations will tend to be given higher weighting if they are closer to the end of the assimilation window. (Tm02 has very similar characteristics to Tm01 and is not discussed further here.)

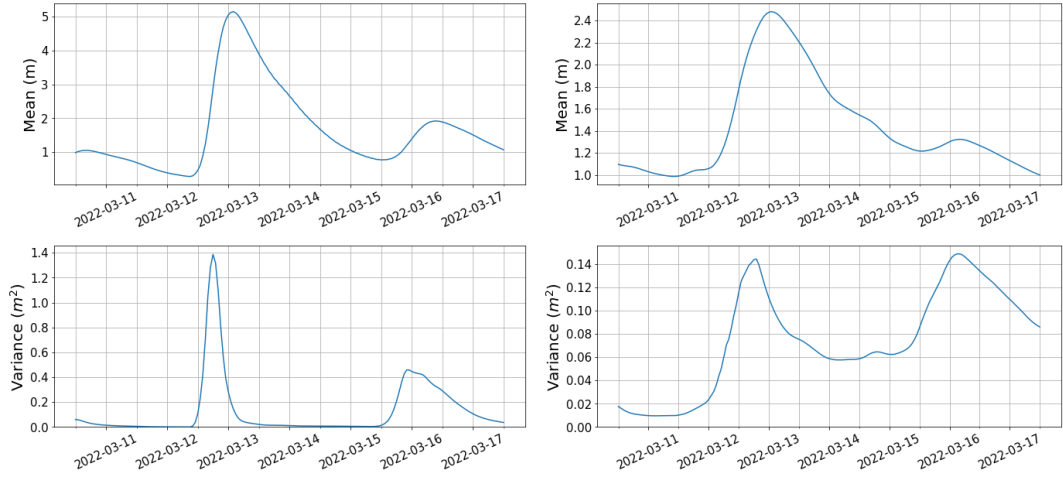


FIGURE 4.1: **Left:** Timeseries of H_s ensemble mean (top) and variance (bottom) from the buoy location. Data is taken from the run beginning 10th March 2022. **Right:** Same as left but a spatial average is taken over the entire domain before taking the ensemble mean and variance.

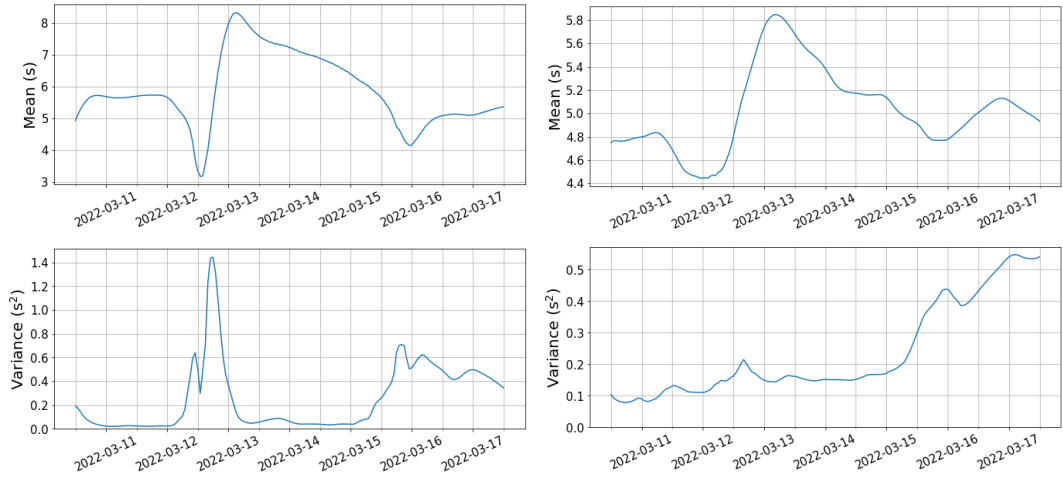


FIGURE 4.2: **Left:** Timeseries of $Tm01$ ensemble mean (top) and variance (bottom) from the buoy location. Data is taken from the run beginning 10th March 2022. **Right:** Same as left but a spatial average is taken over the entire domain before taking the ensemble mean and variance.

4.2 Covariances

4.2.1 Spatial Covariances

4.2.2 Cross Covariances in Time

4.2.3 Cross Covariances Between Variables

Appendix A

Frequently Asked Questions

A.1 How do I change the colors of links?

The color of links can be changed to your liking using:

```
\hypersetup{urlcolor=red}, or
```

```
\hypersetup{citecolor=green}, or
```

```
\hypersetup{allcolor=blue}.
```

If you want to completely hide the links, you can use:

```
\hypersetup{allcolors=.}, or even better:
```

```
\hypersetup{hidelinks}.
```

If you want to have obvious links in the PDF but not the printed text, use:

```
\hypersetup{colorlinks=false}.
```


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